

Evidence Lower Bound (ELBO) in Variational Inference

COM3031 Probabilistic Machine Learning

Setup and Notation

Let $\mathcal{D}$ denote the observed dataset and let θ denote model parameters. We assume a probabilistic model with joint density

$$p(\mathcal{D}, \theta) = p(\mathcal{D} \mid \theta) p(\theta).$$

The posterior is

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D}, \theta)}{p(\mathcal{D})},$$

where the *evidence* (marginal likelihood) is

$$p(\mathcal{D}) = \int p(\mathcal{D}, \theta) d\theta, \quad \log p(\mathcal{D}) = \log \int p(\mathcal{D}, \theta) d\theta.$$

Goal in Variational Inference (as in the slide)

In variational inference, the objective is to approximate the true (typically intractable) posterior $p(\theta \mid \mathcal{D})$ with a tractable distribution $q(\theta)$ from a chosen variational family by minimising the Kullback–Leibler (KL) divergence:

$$\text{KL}(q(\theta) \parallel p(\theta \mid \mathcal{D})) = \int q(\theta) \log \frac{q(\theta)}{p(\theta \mid \mathcal{D})} d\theta.$$

Equivalently, using the definition of expectation under $q(\theta)$,

$$\mathbb{E}_{q(\theta)}[f(\theta)] = \int q(\theta) f(\theta) d\theta.$$

The KL divergence can be written as,

$$\text{KL}(q(\theta) \parallel p(\theta \mid \mathcal{D})) = \mathbb{E}_{q(\theta)}[\log q(\theta) - \log p(\theta \mid \mathcal{D})].$$

However, the posterior $p(\theta \mid \mathcal{D})$ depends on the marginal likelihood $p(\mathcal{D})$, which is typically intractable to compute. As a result, the KL divergence cannot be evaluated directly in this form, motivating the introduction of an alternative objective: the evidence lower bound (ELBO).

From KL to ELBO

Start from the KL definition and substitute Bayes' rule:

$$\begin{aligned}\text{KL}(q(\theta) \parallel p(\theta \mid \mathcal{D})) &= \int q(\theta) \log \frac{q(\theta)}{p(\theta \mid \mathcal{D})} d\theta \\ &= \int q(\theta) \log \frac{q(\theta)}{p(\mathcal{D}, \theta)/p(\mathcal{D})} d\theta \\ &= \int q(\theta) \log \left(\frac{q(\theta) p(\mathcal{D})}{p(\mathcal{D}, \theta)} \right) d\theta \\ &= \int q(\theta) [\log q(\theta) + \log p(\mathcal{D}) - \log p(\mathcal{D}, \theta)] d\theta.\end{aligned}$$

Since $\log p(\mathcal{D})$ does not depend on θ , it can be taken outside the integral:

$$\begin{aligned}\text{KL}(q(\theta) \parallel p(\theta \mid \mathcal{D})) &= \log p(\mathcal{D}) + \int q(\theta) \log q(\theta) d\theta - \int q(\theta) \log p(\mathcal{D}, \theta) d\theta \\ &= \log p(\mathcal{D}) + \mathbb{E}_q[\log q(\theta)] - \mathbb{E}_q[\log p(\mathcal{D}, \theta)].\end{aligned}$$

Rearrange to isolate $\log p(\mathcal{D})$:

$$\log p(\mathcal{D}) = \underbrace{\mathbb{E}_q[\log p(\mathcal{D}, \theta)] - \mathbb{E}_q[\log q(\theta)]}_{\mathcal{L}(q) \text{ (ELBO)}} + \text{KL}(q(\theta) \parallel p(\theta \mid \mathcal{D})).$$

Definition of ELBO

Define the evidence lower bound (ELBO) as

$$\boxed{\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathcal{D}, \theta)] - \mathbb{E}_q[\log q(\theta)] = \mathbb{E}_q \left[\log \frac{p(\mathcal{D}, \theta)}{q(\theta)} \right]}.$$

Because KL divergence is always nonnegative,

$$\text{KL}(q(\theta) \parallel p(\theta \mid \mathcal{D})) \geq 0,$$

we immediately obtain the lower-bound property:

$$\boxed{\mathcal{L}(q) \leq \log p(\mathcal{D}).}$$

This is why $\mathcal{L}(q)$ is called the *evidence lower bound*.

Key Consequence: Optimising ELBO Equals Minimising KL

From the decomposition

$$\log p(\mathcal{D}) = \mathcal{L}(q) + \text{KL}(q(\theta) \parallel p(\theta \mid \mathcal{D})),$$

note that $\log p(\mathcal{D})$ is constant with respect to q . Therefore:

$$\arg \max_q \mathcal{L}(q) \iff \arg \min_q \text{KL}(q(\theta) \parallel p(\theta \mid \mathcal{D})).$$

Moreover, the bound is tight if and only if

$$\text{KL}(q(\theta) \parallel p(\theta \mid \mathcal{D})) = 0 \iff q(\theta) = p(\theta \mid \mathcal{D}).$$

ELBO in the Common “Likelihood + Prior” Form

Using $p(\mathcal{D}, \theta) = p(\mathcal{D} \mid \theta)p(\theta)$, we can expand ELBO:

$$\begin{aligned}\mathcal{L}(q) &= \mathbb{E}_q[\log p(\mathcal{D} \mid \theta)p(\theta)] - \mathbb{E}_q[\log q(\theta)] \\ &= \mathbb{E}_q[\log p(\mathcal{D} \mid \theta)] + \mathbb{E}_q[\log p(\theta)] - \mathbb{E}_q[\log q(\theta)].\end{aligned}$$

Group terms to show the regularisation interpretation:

$$\boxed{\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathcal{D} \mid \theta)] - \text{KL}(q(\theta) \parallel p(\theta)).}$$

So ELBO is:

- an expected log-likelihood term (data fit), minus
- a KL term that penalises deviation from the prior (complexity penalty).

Summary

$$\boxed{\log p(\mathcal{D}) = \mathcal{L}(q) + \text{KL}(q(\theta) \parallel p(\theta \mid \mathcal{D})), \quad \mathcal{L}(q) \leq \log p(\mathcal{D}).}$$

Thus, instead of minimising an intractable KL to the true posterior directly, we maximise the ELBO.